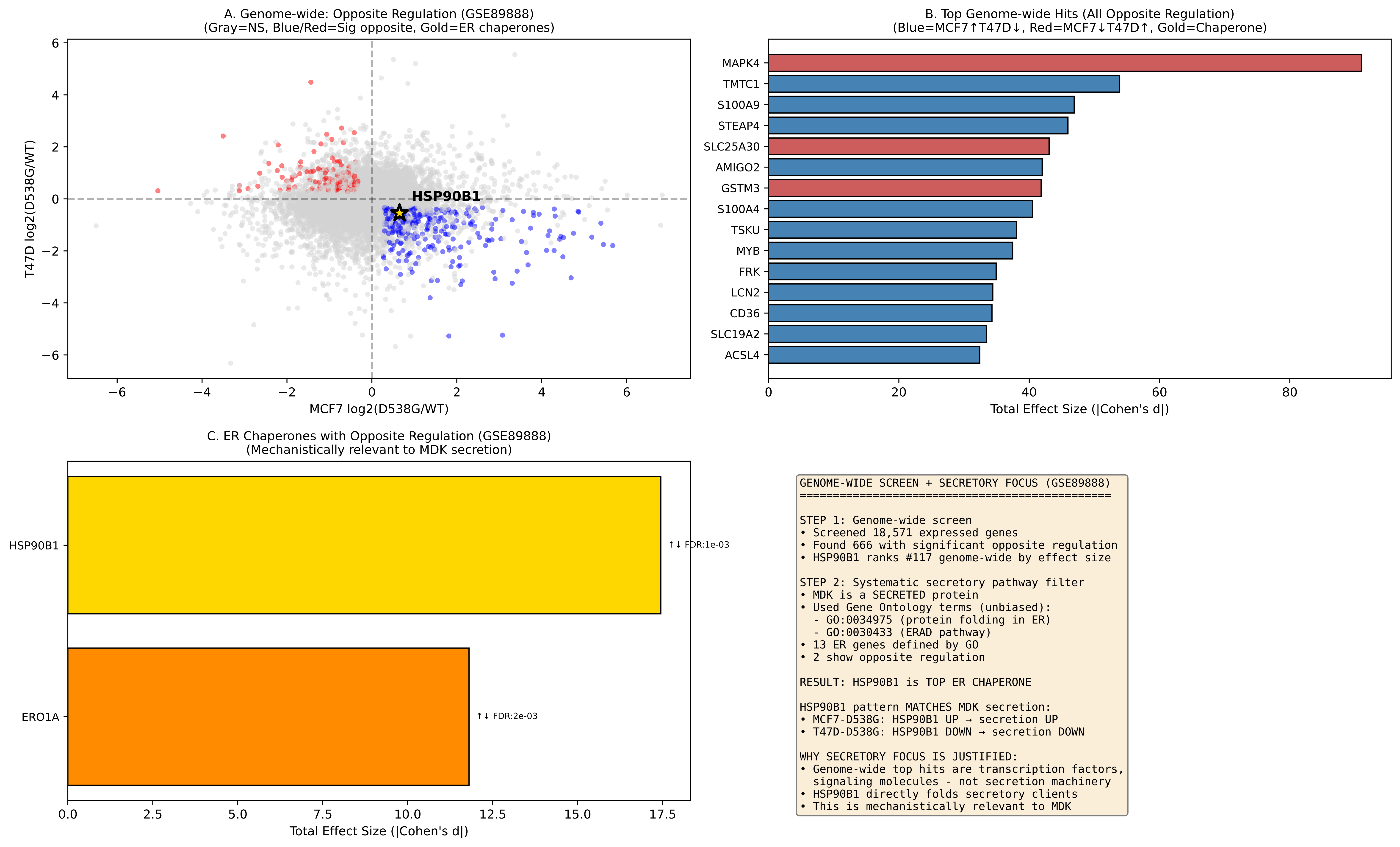

A. Genome-wide: Opposite Regulation (GSE89888)
(Gray=NS, Blue/Red=Sig opposite, Gold=ER chaperones)
HSP90B1
T47D log2(D538G/WT)
MCF7 log2(D538G/WT)
6
4
2
0
−2
−4
−6
B. Top Genome-wide Hits (All Opposite Regulation)
(Blue=MCF7↑T47D↓, Red=MCF7↓T47D↑, Gold=Chaperone)
MAPK4
TMTC1
S100A9
STEAP4
SLC25A30
AMIGO2
GSTM3
S100A4
TSKU
MYB
FRK
LCN2
CD36
SLC19A2
ACSL4
0
20
40
60
80
Total Effect Size (|Cohen's d|)
C. ER Chaperones with Opposite Regulation (GSE89888)
(Mechanistically relevant to MDK secretion)
HSP90B1
↑↓ FDR:1e-03
ERO1A
↑↓ FDR:2e-03
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Total Effect Size (|Cohen's d|)
GENOME-WIDE SCREEN + SECRETORY FOCUS (GSE89888)
===
STEP 1: Genome-wide screen
• Screened 18,571 expressed genes
• Found 666 with significant opposite regulation
• HSP90B1 ranks #117 genome-wide by effect size
STEP 2: Systematic secretory pathway filter
• MDK is a SECRETED protein
• Used Gene Ontology terms (unbiased):
- GO:0034975 (protein folding in ER)
- GO:0030433 (ERAD pathway)
• 13 ER genes defined by GO
• 2 show opposite regulation
RESULT: HSP90B1 is TOP ER CHAPERONE
HSP90B1 pattern MATCHES MDK secretion:
• MCF7-D538G: HSP90B1 UP → secretion UP
• T47D-D538G: HSP90B1 DOWN → secretion DOWN
WHY SECRETORY FOCUS IS JUSTIFIED:
• Genome-wide top hits are transcription factors,
signaling molecules - not secretion machinery
• HSP90B1 directly folds secretory clients
• This is mechanistically relevant to MDK